Trigonometric Ratio: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Ratios and basic identities

Formulas

$$\sin^2\theta+\cos^2\theta=1,\ \sec^2\theta-\tan^2\theta=1,\ \operatorname{cosec}^2\theta-\cot^2\theta=1$$

$$(\sec\theta+\tan\theta)(\sec\theta-\tan\theta)=1,\ (\operatorname{cosec}\theta+\cot\theta)(\operatorname{cosec}\theta-\cot\theta)=1$$

$$\sin^4+\cos^4=1-2\sin^2\cos^2\in[\tfrac{1}{2},1],\ \sin^6+\cos^6=1-3\sin^2\cos^2\in[\tfrac{1}{4},1]$$

$$\frac{\sin^4\theta}{a}+\frac{\cos^4\theta}{b}=\frac{1}{a+b}\Rightarrow\sin^2\theta=\frac{a}{a+b},\ \cos^2\theta=\frac{b}{a+b}$$

KEY POINTS

- $\sin=\frac{P}{H},\ \cos=\frac{B}{H},\ \tan=\frac{P}{B}$; reciprocals $\operatorname{cosec},\sec,\cot$
- Knowing $\sec\theta+\tan\theta$ gives $\sec\theta-\tan\theta$ as its reciprocal
- To eliminate θ: square and add, or use $\sec^2-\tan^2=1$

STEPS

1. Start from the more complicated side
2. Change to sin, cos; take LCM
3. Replace 1 by $\sec^2-\tan^2$ or $\sin^2+\cos^2$ to factorise

2. Angles, quadrants and signs

Formulas

$$\pi\text{ rad}=180^\circ,\quad 1^\circ=\tfrac{\pi}{180}\text{ rad},\quad 1\text{ rad}\approx 57^\circ17'45''$$

KEY POINTS

- No symbol on an angle $\Rightarrow$ radians ($\sin 2$ is $\sin 2$ rad)
- Quadrantal angles: multiples of $\frac{\pi}{2}$; anticlockwise positive
- Signs: Q-I all, Q-II sin, Q-III tan, Q-IV cos (and reciprocals) positive
- $\sin n\pi=0,\ \cos n\pi=(-1)^n$

3. Reduction formulae

Formulas

$$\sin(\tfrac{\pi}{2}-\theta)=\cos\theta,\ \sin(\pi-\theta)=\sin\theta,\ \cos(\pi-\theta)=-\cos\theta,\ \sin(-\theta)=-\sin\theta$$

STEPS

1. Write the angle as $n\frac{\pi}{2}\pm\theta$, θ acute
2. n odd: change sin $\leftrightarrow$ cos, tan $\leftrightarrow$ cot, sec $\leftrightarrow$ cosec; n even: keep
3. Put the sign of the ORIGINAL ratio in that quadrant

KEY POINTS

- Complementary ($A+B=90^\circ$): $\tan A\tan B=1$, $\sin^2A+\sin^2B=1$
- Supplementary ($A+B=180^\circ$): $\sin A=\sin B$, $\cos A=-\cos B$
- $\sin A\pm\cos A=\sqrt{2}\sin(A\pm\frac{\pi}{4})$ decides its sign
- One ratio given: right triangle + Pythagoras + quadrant sign

4. Compound angles

Formulas

$$\sin(A\pm B)=\sin A\cos B\pm\cos A\sin B,\ \cos(A\pm B)=\cos A\cos B\mp\sin A\sin B$$

$$\tan(A\pm B)=\frac{\tan A\pm\tan B}{1\mp\tan A\tan B},\quad \tan(\tfrac{\pi}{4}\pm A)=\frac{1\pm\tan A}{1\mp\tan A}$$

$$\sin(A+B)\sin(A-B)=\sin^2 A-\sin^2 B,\ \cos(A+B)\cos(A-B)=\cos^2 A-\sin^2 B$$

KEY POINTS

- If $A=B+C$: $\tan A-\tan B-\tan C=\tan A\tan B\tan C$
- $\frac{\cos A\pm\sin A}{\cos A\mp\sin A}=\tan(\frac{\pi}{4}\pm A)$

5. Product-sum (C+D) formulae

Formulas

$$2\sin A\cos B=\sin(A+B)+\sin(A-B),\ 2\cos A\cos B=\cos(A+B)+\cos(A-B),\ 2\sin A\sin B=\cos(A-B)-\cos(A+B)$$

$$\sin C\pm\sin D=2\sin\tfrac{C\pm D}{2}\cos\tfrac{C\mp D}{2},\ \cos C+\cos D=2\cos\tfrac{C+D}{2}\cos\tfrac{C-D}{2},\ \cos C-\cos D=-2\sin\tfrac{C+D}{2}\sin\tfrac{C-D}{2}$$

STEPS

1. Pair terms with equal sum or difference of angles
2. Apply $C\pm D$; take the common factor out of numerator and denominator

6. Multiple and sub-multiple angles

Formulas

$$\sin 2A=2\sin A\cos A=\frac{2\tan A}{1+\tan^2 A},\ \cos 2A=1-2\sin^2 A=2\cos^2 A-1=\frac{1-\tan^2 A}{1+\tan^2 A}$$

$$\tan 2A=\frac{2\tan A}{1-\tan^2 A},\quad \tan A=\frac{1-\cos 2A}{\sin 2A}=\frac{\sin 2A}{1+\cos 2A}$$

$$1+\cos A=2\cos^2\tfrac{A}{2},\ 1-\cos A=2\sin^2\tfrac{A}{2},\ 1\pm\sin A=(\cos\tfrac{A}{2}\pm\sin\tfrac{A}{2})^2$$

$$\sin 3A=3\sin A-4\sin^3 A,\ \cos 3A=4\cos^3 A-3\cos A,\ \tan 3A=\frac{3\tan A-\tan^3 A}{1-3\tan^2 A}$$

KEY POINTS

- Half angle: $\sin\frac{A}{2}=\pm\sqrt{\frac{1-\cos A}{2}}$; sign from the quadrant of $\frac{A}{2}$
- $|\sin A\pm\cos A|=\sqrt{1\pm\sin 2A}$
- Powers down: $\sin^3 A=\frac{3\sin A-\sin 3A}{4}$, $\cos^3 A=\frac{3\cos A+\cos 3A}{4}$

Watch out

$\sqrt{1-\cos A}$, $\sqrt{1\pm\sin A}$ are moduli: fix the sign from the interval

7. Standard values

Formulas

$$\sin 15^\circ=\frac{\sqrt{3}-1}{2\sqrt{2}},\ \cos 15^\circ=\frac{\sqrt{3}+1}{2\sqrt{2}},\ \tan 15^\circ=2-\sqrt{3},\ \tan 75^\circ=2+\sqrt{3}$$

$$\tan 22.5^\circ=\sqrt{2}-1,\ \tan 67.5^\circ=\sqrt{2}+1$$

$$\sin 18^\circ=\frac{\sqrt{5}-1}{4},\ \cos 36^\circ=\frac{\sqrt{5}+1}{4},\ \cos 18^\circ=\frac{\sqrt{10+2\sqrt{5}}}{4},\ \sin 36^\circ=\frac{\sqrt{10-2\sqrt{5}}}{4}$$

KEY POINTS

- $\cos 36^\circ-\sin 18^\circ=\frac{1}{2}$, $\cos 36^\circ\sin 18^\circ=\frac{1}{4}$
- $\sin 18^\circ$: root of $4x^2+2x-1=0$; $\sin 36^\circ$: root of $16x^4-20x^2+5=0$

8. Six important results

Formulas

$$\cot A-\tan A=2\cot 2A,\quad \cot A+\tan A=2\operatorname{cosec}2A,\quad (1+\tan A)(1+\tan B)=2\ (A+B=45^\circ)$$

$$\sin\theta\sin(60^\circ-\theta)\sin(60^\circ+\theta)=\tfrac{1}{4}\sin 3\theta,\ \cos\theta\cos(60^\circ-\theta)\cos(60^\circ+\theta)=\tfrac{1}{4}\cos 3\theta$$

$$\tan\theta\tan(60^\circ-\theta)\tan(60^\circ+\theta)=\tan 3\theta$$

KEY POINTS

- Angles $\theta, 60° - \theta, 60° + \theta$ (e.g. $20°, 40°, 80°$) $\Rightarrow$ use the product result
- One factor missing: multiply and divide by it

9. Componendo-dividendo, tan of a sum, continued product

Formulas

$$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

$$\tan(\theta_1 + \cdots + \theta_n) = \frac{S_1 - S_3 + S_5 - \cdots}{1 - S_2 + S_4 - \cdots}$$

$$\cos\theta \cos 2\theta \cdots \cos 2^{n-1}\theta = \frac{\sin 2^n\theta}{2^n \sin\theta}$$

KEY POINTS

- Ratio of sines/cosines given, ratio of tangents asked $\Rightarrow$ componendo-dividendo then $C \pm D$
- Angles doubling $\Rightarrow$ continued product (reorder with $\cos(\pi - x) = -\cos x$ if needed)

10. Identities in a triangle

Formulas

$$\sum \sin 2A = 4\prod \sin A,\ \sum \cos 2A = -1 - 4\prod \cos A,\ \sum \sin^2 A = 2 + 2\prod \cos A,\ \sum \cos^2 A = 1 - 2\prod \cos A$$

$$\sum \tan A = \prod \tan A,\ \sum \cot A \cot B = 1,\ \sum \tan\tfrac{A}{2} \tan\tfrac{B}{2} = 1,\ \sum \cot\tfrac{A}{2} = \prod \cot\tfrac{A}{2}$$

KEY POINTS

- Use $A + B = \pi - C$: $\sin(A+B) = \sin C$, $\cos(A+B) = -\cos C$, $\sin\frac{A+B}{2} = \cos\frac{C}{2}$
- $\sum \sin^2 A = 2$ or $\sum \cos^2 A = 1 \Rightarrow$ right-angled triangle

11. Graphs and ranges (types 1, 2)

Formulas

$$\sin, \cos \in [-1, 1];\ \tan, \cot \in \mathbb{R};\ \sec, \operatorname{cosec} \in (-\infty, -1] \cup [1, \infty)$$

KEY POINTS

- Periods: $\sin, \cos, \sec, \operatorname{cosec}$: 2π; $\tan, \cot$: π; $|\sin x| + |\cos x|$: $\frac{\pi}{2}$
- Restricted x: read the range from the graph on that interval
- $af + b$: scale and shift (ends swap if $a < 0$); f^2, $\sqrt{f}$, $\frac{1}{f}$: watch where $f = 0$
- Quadratic in one trig function: complete the square, $t \in [-1, 1]$, vertex or end-points

12. Ranges (types 3, 4)

Formulas

$$a\sin x + b\cos x \in [-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}], \quad |\sin x| + |\cos x| \in [1, \sqrt{2}]$$

$$a^2\tan^2 x + b^2\cot^2 x \geq 2ab, \quad a^2\sec^2 x + b^2\operatorname{cosec}^2 x \geq (a+b)^2$$

KEY POINTS

- $a\sin^2 x + b\cos^2 x + c\sin x\cos x$: change to $2x$, then type 3
- Restricted x: write $R\sin(x+\phi)$ and use the interval of $x + \phi$
- $a^2\sin^2 x + b^2\operatorname{cosec}^2 x$: if $\sin^2 x = \frac{b}{a} \leq 1$ min $2ab$, else min $a^2 + b^2$ (same for $\cos^2, \sec^2$)

Watch out

Add two ranges only if both extremes occur at the same x

13. Summation of series

Formulas

$$\sum_{k=0}^{n-1} \sin(\alpha + k\beta) = \frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}} \sin\left(\alpha + \frac{(n-1)\beta}{2}\right) \text{ (cos likewise), } \beta \neq 2k\pi$$

$$\frac{\sin(A-B)}{\cos A \cos B} = \tan A - \tan B, \quad \frac{\sin(A-B)}{\sin A \sin B} = \cot B - \cot A, \quad \operatorname{cosec} 2x = \cot x - \cot 2x$$

STEPS

1. Angles in A.P.: type 1 formula (squares/cubes: reduce the power first)
2. Product of two trig terms in the denominator: multiply by sin of their difference
3. Write $T_r = f(r+1) - f(r)$; sum $= f(n+1) - f(1)$

14. Triangle inequalities and substitution

Formulas

$$\sum \cos A \leq \tfrac{3}{2}, \quad \sum \sin A \leq \tfrac{3\sqrt{3}}{2}, \quad \prod \cos A \leq \tfrac{1}{8}, \quad \sum \cot^2 A \geq 1, \quad \sum \tan^2 \tfrac{A}{2} \geq 1$$

KEY POINTS

- Symmetric expression: extreme at $A = B = C = 60°$; compare with a degenerate triangle to decide max or min
- $x^2 + y^2 = r^2$: $x = r\cos\theta$, $y = r\sin\theta$; $1 + x^2$: $x = \tan\theta$
- Choose the parameter interval so each value appears once

15. Common mistakes checklist

STEPS

1. Angle without a degree sign is in radians
2. Reduction: sign of the original ratio, not of the new one
3. Square roots give moduli: $\sqrt{\sin^2 x} = |\sin x|$
4. Half angle sign from the quadrant of $\frac{A}{2}$, not of A
5. Range: check that the extreme value is actually attained
6. Series type 1: n = number of terms, β = common difference